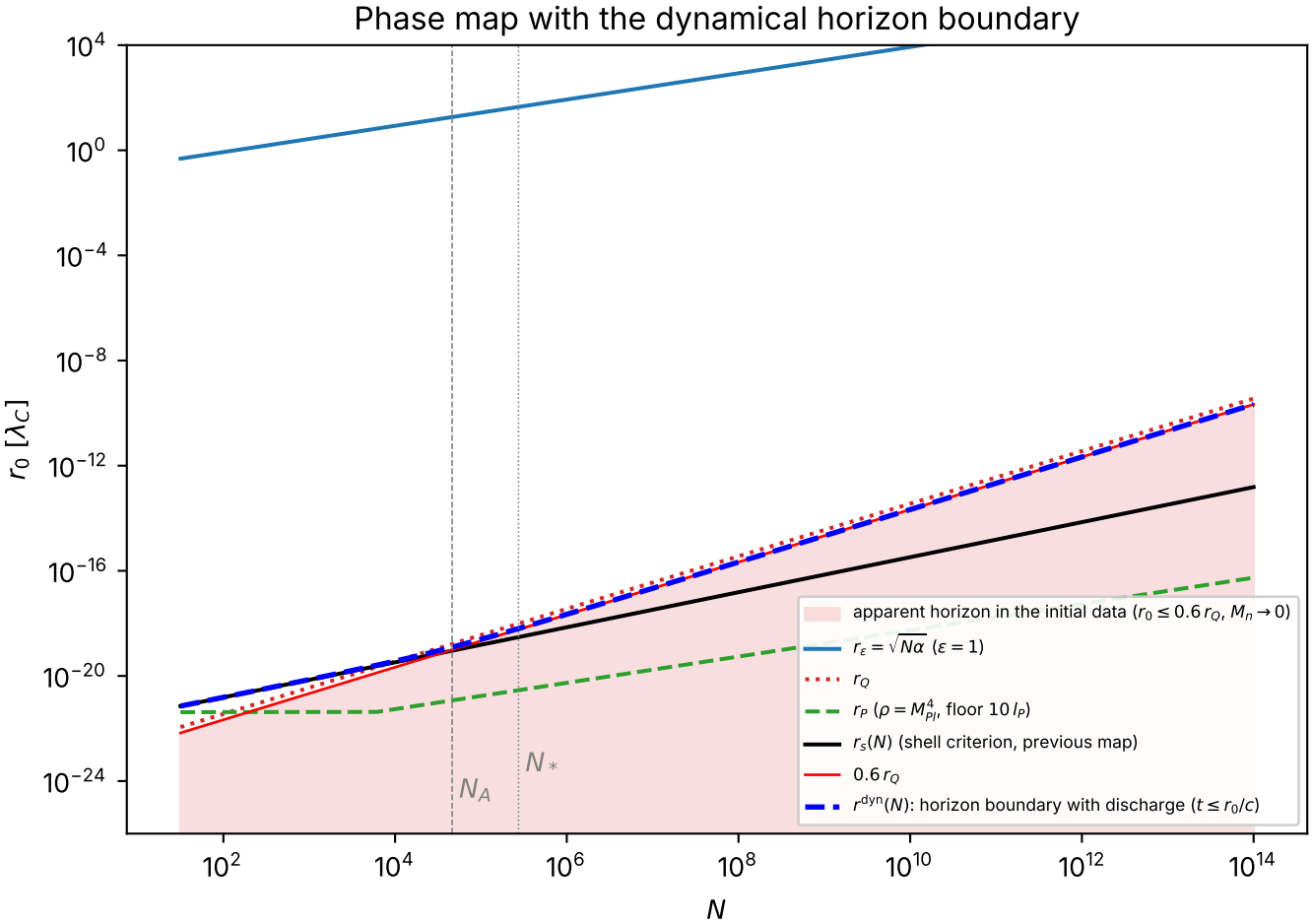
Phase map with the dynamical horizon boundary
r_0 [λ_C]
N
10^{4}
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{-20}
10^{-24}
10^{2}
10^{4}
10^{6}
10^{8}
10^{10}
10^{12}
10^{14}
N_A
N_*
apparent horizon in the initial data ($r_0 \le 0.6\, r_Q$, $M_n \to 0$)
$r_\varepsilon = \sqrt{N\alpha}$ ($\varepsilon = 1$)
r_Q
r_P ($\rho = M_{Pl}^4$, floor $10\, l_P$)
$r_s(N)$ (shell criterion, previous map)
$0.6\, r_Q$
$r^{\mathrm{dyn}}(N)$: horizon boundary with discharge ($t \le r_0/c$)